

SVBT is a measurement lens for detecting irreversible contraction of future-generative capacity outside an agent's current evaluator.

Future-generative capacity remaining after action $\mathbf{a}$

$$\Gamma_t(a) = \bigcup_{z \in \text{Reach}_F(z_a)} \text{Out}(z)$$

Generative capacity lost relative to a reference action $\mathbf{b}$

$$L_t(a \mid b) = \Gamma_t(b) \setminus \Gamma_t(a)$$

Irreversible loss of generative conditions

$$\text{Reach}_F(z_a) \cap C = \emptyset$$

After action $\mathbf{a}$, the system can no longer reach any configuration in which the required generative condition C exists.

SVBT does not assign utility to unknown future possibilities. It asks whether the conditions that could generate them have been irreversibly removed.